Advances in Data Analysis and Statistical Modelling

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1 Introduction

This project focuses on analyzing the Quality of Life 2024 dataset, which collects information on 178 cities and their respective quality-of-life indicators. The goal of the analysis is twofold: first, to apply standardization, distance measurements, and the computation of aggregate indices; second, to use advanced dimensionality reduction and clustering techniques to uncover patterns in the data and evaluate the performance of different methodologies.

In this case, a random subset of 60 cities was selected from the dataset to perform fundamental data analysis techniques.

The dataset contains information about 60 cities, each characterized by several indices:

- Quality of Life Index
- Purchasing Power Index
- Safety Index
- Health Care Index
- Cost of Living Index
- Property Price to Income Ratio
- Traffic Commute Time Index
- Pollution Index and Climate Index.

1.1 Standardize data

The first step is to standardize data that is crucial because standardization eliminates bias due to differing measurement units among indices. In this case, we use the centering matrix Jc to centralize the dataset around its mean.

Then we calculate the covariance matrix Sc of the original data and extract the square root of the diagonal of Sc (standard deviations), so we obtain the standardized matrix Z, ensuring variables are on the same scale. Therefore the variance-covariance matrix of the standardized data Z equals the correlation matrix Sz of the original dataset.

In table 1, we can note that there is a strong positive correlations between Quality of Life Index and Purchasing Power Index (0.8896), this means that countries with higher purchasing power tend to have a better quality of life. The relation between Quality of Life index and Climate index is equals to 0.7453, suggests that a good climate is often associated with a higher quality of life. We can also observe that countries with higher purchasing power tend to have favorable climates.

In contrast, we can note that there is a negative correlation between Quality of Life index and Pollution index (-0.8861) indicates that high pollution levels drastically lower quality of life. The correlation between purchasing power index and pollution index is also negative (-0.6860) so countries with higher purchasing power tend to have better pollution control.

	QLI	**PPI**	**SI**	**HCI**	**CLI**	**PPIR**	**TCTI**	**PI**	**CI**
QLI	1.0000	0.8896	0.6840	0.6631	0.7453	-0.6389	-0.6806	-0.8861	0.0054
PPI	0.8896	1.0000	0.5599	0.5138	0.7518	-0.6159	-0.4365	-0.6860	-0.1006
SI	0.6840	0.5599	1.0000	0.4763	0.3188	-0.2492	-0.6246	-0.5282	-0.2600
HCI	0.6631	0.5138	0.4763	1.0000	0.4497	-0.2064	-0.4538	-0.5811	0.0779
CLI	0.7453	0.7518	0.3188	0.4497	1.0000	-0.4258	-0.4232	-0.7218	0.1108
PPIR	-0.6389	-0.6159	-0.2492	-0.2064	-0.4258	1.0000	0.3416	0.4804	0.1704
TCTI	-0.6806	-0.4365	-0.6246	-0.4538	-0.4232	0.3416	1.0000	0.5970	0.0432
PI	-0.8861	-0.6860	-0.5282	-0.5811	-0.7218	0.4804	0.5970	1.0000	0.0559
CI	0.0054	-0.1006	-0.2600	0.0779	0.1108	0.1704	0.0432	0.0559	1.0000

Table 1: Matrix of correlation coefficients

1.2 The Euclidean distance between units

In the second step we compute the pairwise Euclidean distances between rows of the standardized matrix Z. The Euclidean distance measures similarity between cities based on the standardized indices, where smaller distances imply higher similarity, while larger distances indicate greater dissimilarity.

1.3 Well-structured perfect partition

The most parsimonious model for describing a clustering by using dissimilarities assumes equal heterogeneity for each class of C and equal isolation between pairs of C classes. Suppose that value α_1 measures the heterogeneity of the objects within each class of C; while α_2 evaluates the isolation between classes. Of course,it is suitable that $\alpha_1 \leq \alpha_2$, because this implies that a function of the within cluster dissimilarities is smaller than a function of the between cluster dissimilarities. The semiparametric well-structured perfect classification matrix P is given by:

$$D_c = P = \alpha_2 \left(1_n 1_n' - UU'\right) + \alpha_1 (UU' - I_n)$$

where 1_n is a vector of n ones, I_n is the identity matrix of order n and matrix U is a $(n \times K)$ membership matrix, binary and row-stochastic. In fact, $m_{ik} = 1$ if the ith object o_i belongs to the kth class C_k , $m_{ik} = 0$ otherwise. In this model all the within clusters dissimilarities are supposed equal to α_1 while all the between clusters dissimilarities are hypothesized equal to α_2. Thus, P is characterized by (U, α_1, α_2).

In this dataset, first we compute the WSPP for clustering the data into 2 clusters and we obtain that:

- $\alpha_1 = 3.40$: the heterogeneity within the clusters is moderate, so the clusters are not perfectly homogeneous but are not overly different either.
- $\alpha_2 = 5.68$: the isolation between the clusters is quite strong, indicating that the clusters are well-separated from each other.

After obtaining the results, we compute the pF value, which likely measures the quality of the solution. Where pF_2 indicates the performance metric for the class with $k = 2$, that is equal to 24.4670.

Then we compute the same procedure for $k = 3$ and $k = 4$ clusters. Respectively we get the correspondents pF.

pF_2	24.4670
pF_3	13.7053
pF_4	22.3239

Table 2: pF values WSPP

These values likely indicate the quality or fitness of the clustering solutions, where a higher pF value is better. The clustering solution for $k = 2$ yields the highest pF value, which suggests that the two-cluster solution provides the best grouping in terms of minimizing the distances within clusters and maximizing the distances between clusters. This result implies that, at least for this dataset, a division into two clusters is optimal, providing a more distinct separation of the data points.

The pF value drops when increasing the number of clusters to 3. This suggests that the three-cluster solution does not improve the clustering quality compared to the two-cluster solution. In fact, the performance of the solution worsens, meaning that the data might not naturally separate into three clusters, and introducing an additional cluster leads to unnecessary complexity.

While the pF value for 4 clusters is better than for 3, it is still lower than for 2 clusters. This indicates that increasing the number of clusters to 4 does not lead to a better solution than 2 clusters.

1.4 Well-structured partition

Let us now relax the hypothesis of equal heterogeneities for all classes and equal isolations between classes of the partition. A second more flexible classification matrix allows for specifying different heterogeneities and isolations. Given a partition $C = \{C_1, \ldots, C_k, \ldots, C_K\}$ of O such that objects within C_k have heterogeneity ${}_{\mathrm{W}}d_{kk} > 0\, \forall k$, while objects between C_k and C_h have isolation ${}_{\mathrm{B}}d_{kh} > 0\, \forall k, h\, (k \neq h)$, a dissimilarity classification matrix Q, defined as a function of the square matrices of order K, $D_B = \left[{}_{\mathrm{B}}d_{kh} > 0 : {}_{\mathrm{B}}d_{kk} = 0, h, k = 1, \ldots, K\, (k \neq h)\right]$, $D_W = \left[{}_{\mathrm{W}}d_{kk} > 0 : {}_{\mathrm{W}}d_{kh} = 0, h, k = 1, \ldots, K\, (k \neq h)\right]$, and a membership matrix U can be one-to-one associated to C. The semiparametric well-structured classification matrix Q has the form,

$$D_c = Q = UD_BU' + UD_WU' - diag(UD_WU')$$

Therefore, the classification matrix Q specifies a more flexible kind of partition, where both the expected clusters heterogeneities and the expected between clusters isolations may differ.

In this case, as the number of clusters increases (from 2 to 3 and 4), D_b increases, meaning the clusters are becoming more distinct or separated from each other. This suggests that dividing the countries into more clusters results in higher dissimilarity, as expected. We can note that $D_b2 = 160.21$ is the lowest, indicating that the two clusters have the least separation. As we increase the number of clusters to 3 and 4, the separation increases, with $D_b4 = 291.48$ showing the largest dissimilarity. However, this doesn't necessarily imply the best clustering structure as we need to consider the other factors (like D_w and pF) to evaluate the clustering quality.

As the number of clusters increases, D_w decreases, which indicates that the homogeneity within each cluster improves as the number of clusters grows. For example, $D_w2 = 379.78$ suggests that with two clusters, there is a larger within-cluster dissimilarity compared to $D_w4 = 248.51$, where the dissimilarity is smaller and the clusters are more internally cohesive. This suggests that increasing the number of clusters leads to better internal homogeneity, which is generally a good outcome for clustering.

pF_2	24.4670
pF_3	29.0063
pF_4	21.8944

Table 3: pF values WSP

The best clustering solution, according to pF, is achieved with 3 clusters (pF_3 = 29.00). This suggests a balance between having sufficient separation between clusters (D_b) and cohesiveness within clusters (D_w). Increasing the number of clusters to 4 seems to reduce the partition's performance, as indicated by the decrease in pF_4. Therefore, 3 clusters are likely the most appropriate solution for the data, balancing separation between groups and internal homogeneity.

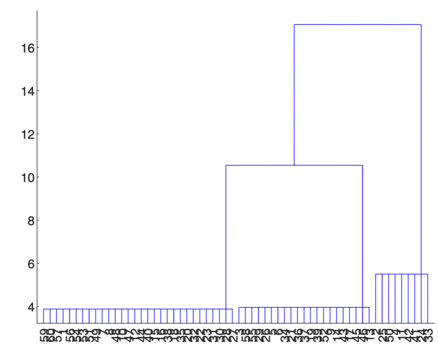


Figure 1: Parsimonious Dendogram

1.5 Parsimonious Dendogram

The term ”parsimonious dendrogram” refers to the creation of a hierarchical tree diagram (dendrogram) based on clustering results, but with a focus on simplicity and efficiency in grouping similar data points into clusters. A parsimonious dendrogram is designed to minimize complexity while still capturing the underlying structure of the data, which can be useful in understanding the relationships among countries based on our indices.

In our code, we applied the PD algorithm with different numbers of clusters (2, 3, 4, 5), and we obtain their respective pF values. The highest pf value corresponds to $k = 3$ clusters with $pF3 = 29.0063$. This suggests that the partitioning with 3 clusters best captures the balance between internal homogeneity (D_w) and between cluster isolation (D_b), leading to a better defined clustering structure. As we increase the number of clusters (to 4 and 5), the pF values decrease, indicating that additional clusters might be overfitting the data or failing to produce more meaningful or distinct partitions. The higher number of clusters introduces more complexity without providing substantial improvement in the clustering quality.

Moreover, in Figure 1, we can observe that three primary branches are visible, representing the 3 optimal clusters. Each branch groups similar elements together. The horizontal lines in the dendrogram indicate the similarity or distance at which clusters are merged. Longer horizontal lines represent larger distances (less similarity) between the merged clusters.

We can note that cluster 1 contains countries that are significantly different from the other two clusters (as indicated by the high between-cluster distances). However, the countries within cluster 1 show greater diversity among themselves. Clusters 2 and 3 are closer to each other, suggesting that they share some similarities. Cluster 3 is the most homogeneous, while cluster 2 shows slightly more variability. In summary, the partitioning with 3 clusters (based on $pF_3 = 29.0063$) would be the most appropriate choice based on the results we have obtained.

At the end of the analysis, we verify if the Q matrices are ultrametric. The ultrametric property states that for any three objects,

$$d(a, c) \leq \max(d(a, b), d(b, c))$$

This property strengthens the triangle inequality by requiring that the longest pairwise distance must always dominate.

Q_2	0
Q_3	0
Q_4	25.9016
Q_5	160.2710

Table 4: verify ultrametric

These matrices Q_2 and Q_3 perfectly satisfy the ultrametric condition. A result of 0 indicates no violation of the ultrametric property, making them valid representations for hierarchical clustering and dendrogram construction.

2 Tandem Analysis

2.1 Principal Component Analysis

Given a matrix X of data relating to n statistical units on which J has been observed quantitative variables, the general purpose of the Principal Component Analysis (PCA) is to construct analytically a new data matrix Y, not directly observable, consisting of new quantitative variables not collected, called principal components, which are generally less or equal to J. The principal components in the columns of the Y matrix are linear combinations of the J variables in the X matrix. The new variables are constructed so that they are in pairs and such that they seek the maximum total variance of X if they are less than J, otherwise they equalize this variance.

In this case, after compute the algorithm of PCA, we obtain the matrix of eigenvectors A and the diagonal matrix of eigenvalues L(variance explained by each principal component). The goal is to identify components with eigenvalues greater than 1, which contribute significantly to the variance in the data. Only the first two principal components have eigenvalues greater than 1, but as we can note in table 5, also the third eigenvalue is very close to 1. However, following the rule of the eigenvalue greater than 1, we choose 2 principal components. These two components explain the majority of the variance in the data, simplifying the dataset while preserving essential information.

From the matrix of eigenvalues it is observed that the first two eigenvalues 1+2=5.0145+1.1898=6.2044 explain 6.2044/9 * 100 $\approx$ 70% of the total variance of the X matrix.

We rotate the components to improve interpretability without altering the variance structure. After rotation, the loadings indicate stronger relationships between certain indices and components.

5.0145	0	0	0	0	0	0	0	0
0	1.1898	0	0	0	0	0	0	0
0	0	0.9824	0	0	0	0	0	0
0	0	0	0.5833	0	0	0	0	0
0	0	0	0	0.4653	0	0	0	0
0	0	0	0	0	0.3980	0	0	0
0	0	0	0	0	0	0.2416	0	0
0	0	0	0	0	0	0	0.1247	0
0	0	0	0	0	0	0	0	0.0000

Table 5: the diagonal matrix of eigenvalues

Label	**Factor 1**	**Factor 2**
QLI	-0.4047	0.1849
PPI	-0.3310	0.2100
SI	-0.0941	0.4476
HCI	-0.3628	-0.0043
CLI	-0.4468	-0.0614
PPIR	0.1505	-0.2866
TCTI	0.2278	-0.2412
PI	0.3699	-0.1515
CI	-0.4185	-0.7445

Table 6: Rotated Factors

In particular, we note that the pollution index is positively correlate with Factor 1 (0.3699) and also traffic commute time and property price to income ratio show weaker positive contributions. At the same time, we observe that the cost of living index is negatively correlate (-0.4468) with Factor 1 and climate index also has a strong negative correlation (-0.4185) with Factor 1.

In contrast, we note that safety index has a strong positive correlation (0.4476) with Factor 2 and purchasing power and quality of life index also positively correlate, but less strongly. But climate index is strongly negatively correlate with Factor 2 and property price to income ratio and traffic commute time index also have weaker negative contributions.

Therefore Factor 1 seems to reflect a general living condition gradient, where high living costs and poor climate contribute negatively, while higher pollution and traffic-related variables align positively. However Factor 2 may represent safety and environmental quality, where high safety correlates positively and poorer climate or affordability conditions negatively influence this factor.

2.2 K-means

In this section, we apply the algorithm called K-means, which is one of the most popular iterative descent clustering methods. The criterion is minimized

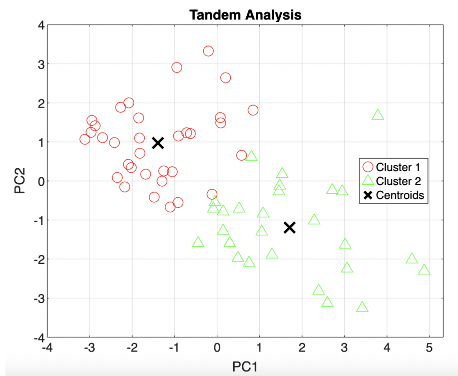


Figure 2: Tandem Analysis. k-means classification computed on the first two principal components.

by assigning the N observations to the K clusters in such a way that within each cluster the average dissimilarity of the observations from the cluster mean, as defined by the points in that cluster, is minimized.

Thanks to the tandem model, on the factorial scores Y, a K-means clustering is applied in order to obtain a dimensional reduction (clustering) of the units.

We replicate the clustering 50 times for each number of clusters with random starting points in order to find the best possible solutions, that are the ones with the lowest within cluster sums of point to centroid distances. The output of the procedure is, for each clustering, a vector of cluster indices of each observation, and the centroids locations.

In order to choose the number of clusters to consider, compute the Pseudo-F statistic for each of the three possible number of clusters. The Pseudo-F statistic is the ratio between the variance between clusters and the variance within clusters, each of them weighted with the corresponding degrees of freedom. To compute the pseudo-F statistic, we compute the cluster membership matrices for each of the two partitions found.

In this case, we note that the psF value for K=2 is significantly higher than K=3, indicating better clustering with two clusters. This suggests that the clusters are more compact and well-separated for K=2.

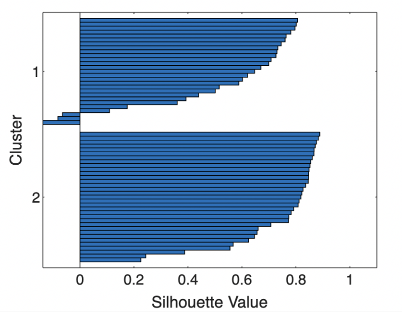
	K = 2	**K = 3**
psF	77.1341	70.2541
Mean Silhouette	0.6487	0.5878

Table 7: Comparison of clustering metrics for $K = 2$ and $K = 3$.

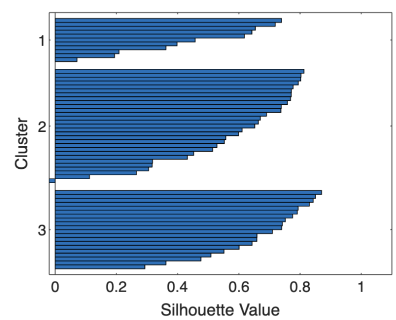
Then we analyzed the partition Silhouette F-statistic (psF) that ensures the compactness (within-cluster scatter,Dw) and separation (between-cluster scatter, Db), so a higher psF indicates better clustering. The silhouette value evaluates the quality of clustering for each point, ranging from -1 to 1, where:

- Values > 0.6 indicate good clustering.
- Values close to 0 suggest points are on cluster boundaries.
- Negative values imply misclassified points.

In this case we can note that the mean silhouette value for $K = 2$ is 0.6487, which exceeds the 0.6 threshold for good clustering. In contrast, the value for $K = 3$ is lower, at 0.5878, indicating slightly poorer cluster cohesion and separation.



(a) Figure 1



(b) Figure 2

The silhouette plots confirm the results:

- For $K = 2$, (Figure 1) the majority of data points have silhouette values above 0.6, indicating clear and well-separated clusters.
- For $K = 3$, (Figure 2) there is a noticeable drop in silhouette values, with some points having values below 0.6, indicating weaker cluster definitions and increased ambiguity.

Based on the explained variance and the mean silhouette score, $K = 2$ is the optimal choice. It captures a higher proportion of the variance in the data and demonstrates better clustering quality. While $K = 3$ provides additional clusters, the trade-off is reduced explained variance and less cohesive clustering, making $K = 2$ the preferred solution.

3 Reduced K-means

The Reduced k-means clustering is a method for clustering objects in a low-dimensional subspace. The advantage of this method is that both clustering of objects and low-dimensional subspace reflecting the cluster structure are simultaneously obtained. In Reduced k-means clustering, the numbers of clusters and dimensions, k and q, have to be appropriately determined such that the cluster result can be optimized.

Figure 3: Reduced k-means

Therefore, we applied the Reduced k-means algorithm in our dataset for both 2 and 3 clusters. We can observe that the 2 cluster solution has an explained variance (39.76%) close to the 3 cluster solution (39.73%), but the pF value of k=2 is higher than k=3 that indicates better cluster compactness and separation.

In figure 3, we observe that the clusters appear to be well-separated in the reduced feature space, indicating that the Reduced K-Means algorithm successfully grouped the data based on meaningful patterns. The boundary between the clusters is not explicitly shown, but the spatial distribution suggests a clear distinction.

It can be observed that adding more clusters (K=3) increases complexity without a corresponding increase in the ability to explain the data structure.

	$K = 2$	$K = 3$
Percentage Explained Variance	39.7653%	39.7375%
p_F	121.4742	104.7710

Table 8: Comparison of metrics for $K = 2$ and $K = 3$.

4 Factorial k-means

The procedure combines k-means cluster analysis with aspects of factor analysis and PCA and is, therefore, called factorial k-means analysis. It is a discrete and continuous model that simultaneously classifies objects and finds a subset of factors that best describe the classification. The methodology allows one to select the most relevant components for the classification. A representation in a reduced number of dimensions can be given to help the interpretation of relationships within the set of variables and the set of objects.

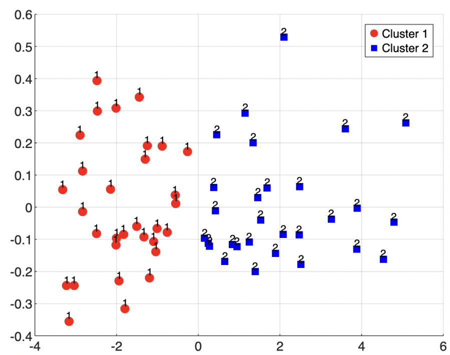


Figure 4: Classification of 60 countries of first two dimension of factorial k-means

This methodology is a valid alternative to the frequently used tandem analysis. With respect to this last sequential approach of factorial and classification analysis, the factorial k-means analysis identifies main factors and the partition of the objects simultaneously by minimizing a single objective function. By contrast, tandem analysis produces factors and an optimal classification, thus minimizing two different objective functions that may work in contradiction, identifying factors that do not contribute much to perceive the clustering structure, and that conversely, may in part mask it.

In this case we can visualize on the first two dimensions of the factorial space the cluster structure. The clusters are well-separated along the two plotted dimensions, indicating effective classification. Cluster 1 is concentrated on the left side of the graph, with a spread primarily along the vertical axis, rather cluster 2 is located on the right side, with its points distributed mostly along the horizontal axis.

The percentage of explained variance for factorial k-means is 39.7375%, which means that these two dimensions capture approximately 40% of the total variability in the dataset. Therefore the clusters appear distinct and show minimal overlap, indicating that the classification achieved by the Factorial K-means is effective.

	$K = 2$	$K = 3$
p_F	146.8649	106.5129

Table 9: Comparison of p_F for $K = 2$ and $K = 3$.

Comparing the two graphical representations in Figs. 2 and 4 , it can be observed that the factorial k-means analysis, more clearly shows the two classes detected by the classification on the variables than the tandem analysis.

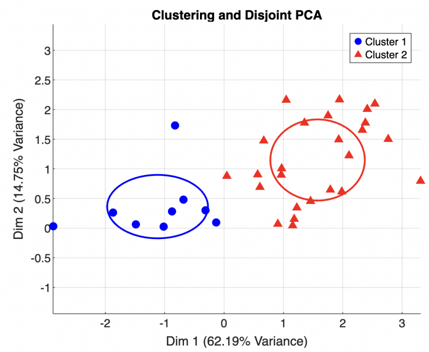


Figure 5: Clustering and Disjoint PCA

5 Clustering and Disjoint PCA

This method is a constrained principal component analysis, which aims at a simultaneous clustering of objects and a partitioning of variables. The new methodology allows us to identify components with maximum variance, each one a linear combination of a subset of variables. All the subsets form a partition of variables. Simultaneously, a partition of objects is also computed maximizing the between cluster variance.

In this case, we compute the CDPCA algorithm and we observe that the percentage explained variance (39.6105%) is slightly lower than other methods. This suggests that clustering and disjoint PCA might not capture the variance as effectively in this dataset. Compared to other clustering techniques like factorial k-means and reduced k-means, CDPCA emphasizes local PCA within clusters. While this may result in slightly lower explained variance, it can provide better insights into cluster-specific structures.

The results of the CDPCA are reported in Fig.4, by considering a biplot representation. The two clusters of countries are also highlighted by two dotted ellipses, while the two clusters of variables are represented by the two dotted orthogonal axes.

6 Double k-means

The model can identify both different classification structures of objects and variables (e.g., partitions, coverings) and different classification types (hard or fuzzy). The identification of the model is obtained by the numerical solution of a least-squares problem. The model is named double k-means since its features recall for some extent the well-known clustering k-means methodology. The difference between the CDPCA and Double k-means models is that in Double K-Means we don't give different weights to the variables in the same cluster. In this case, we compute the algorithm considering 2 clusters and the explained variance is equal 37.03%, so the first two components of the PCA capture around

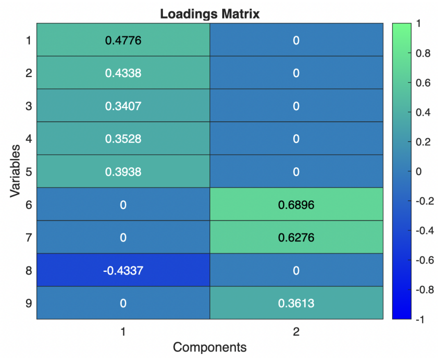


Figure 6: Loadings Matrix CDPCA

37.07% of the overall variability in the dataset.

7 Compare the results by using the confusion matrix

We compare, for every couple of model that we used in previous sections, the partitions of units obtained using confusion matrices.

We considered the confusion matrix which represents the alignment between two cluster assignments: one derived from the k-means (U2) and the other from the Reduced K-means clustering (Urkm).

$$\text{Confusion Matrix: } U2' \times Urkm = \begin{pmatrix} 32 & 0 \\ 3 & 24 \end{pmatrix} \tag{1}$$

The rows represent clusters from the k-means (U2) and the columns represent clusters from reduced k-means (Urkm). We can note that 33 countries from cluster 1 in k-means are correctly assigned to cluster 1 in reduced k-means and 24 countries from cluster 2 in U2 are also correctly assigned to cluster 2 in Urkm. So the majority of the countries are correctly classified by the reduced k-mean clustering method, as shown by the high values on the diagonal (33 and 24). Despite there are only 3 countries that are misclassified , and all are from cluster 2 in k-means being assigned to cluster 1 in reduced k-means. This suggests that the clustering structure of reduced k-means aligns well with k-means.

Then we compute the confusion matrix between k-means and factorial k-means. We note that out of a total of 60 countries, 31+27=58 countries were correctly classified and the total number of misclassifications is 2, all occurring in Cluster 2.

$$\text{Confusion Matrix} = \begin{bmatrix} 31 & 2 \\ 0 & 27 \end{bmatrix} \tag{2}$$

Computing the confusion matrix between k-means and CDPCA, we note that CDPCA correctly classified 33 countries as belonging to Cluster 1. There were no misclassifications of countries from Cluster 1 into Cluster 2 by CDPCA. This indicates a high agreement between K-means and CDPCA for Cluster 1. Therefore we observe that CDPCA correctly classified 25 countries as belonging to Cluster 2. However, 2 countries from Cluster 2 were misclassified by CDPCA as belonging to Cluster 1.

$$\text{Confusion Matrix} = \begin{bmatrix} 32 & 2 \\ 0 & 25 \end{bmatrix} \tag{3}$$

All three methods (Reduced K-means, Factorial K-means, and CDPCA) demonstrate strong alignment with the K-means clustering structure, as evidenced by the high diagonal values in the confusion matrices. Reduced K-means and Factorial K-means slightly outperform CDPCA in terms of overall accuracy, with fewer misclassifications. CDPCA's performance is particularly strong for Cluster 1, with no misclassifications, but less consistent for Cluster 2.

8 The explained variance of the different methods

Reduced K-mean	39.7653 %
Factorial K-means	39.7375%
Clustering and Disjoint PCA	39.6105 %
Double K-means	37.0273 %

Table 10: The explained variance of the different methods

Comparing the results we note that models like Reduced K-means and Factorial K-means are designed to optimize explained variance, making them suitable for applications where preserving the data structure is critical. CDPCA, on the other hand, prioritizes interpretability by enforcing sparsity in the cluster assignments, which explains its slightly lower variance capture.

CDPCA sacrifices a small amount of explained variance to gain interpretability and sparsity, which is ideal for high-dimensional datasets where understanding the role of individual variables is important. However, Double K-means, while simpler, may not be the best choice when retaining data variability is crucial, as it explains significantly less variance.

Moreover we can conclude that if variance explained is the primary objective, Reduced K-means or Factorial K-means are preferable. If the interpretability is more important, the CDPCA is a strong candidate.

9 Matlab code

```
QLCit24
rng(09042002)
```

```
rowp = randperm(178);
Xp=X(rowp, :);
Xr=Xp(1:60,:);

%standarize data
n=size(Xr,1)
J=size(Xr,2)
Jc=eye(n,n)-(1/n)*ones(n,n)
%coveriance matrix
Sc=(1/60)*Xr'*Jc*Xr;
D=diag(diag(Sc).^0.5);
Z=Jc*Xr*D^(-1);
%correlation matrix
Sz=(1/60)*Z'*Z;

%2. Cdistv=pdist(Z,"euclidean");
distv=pdist(Z,"euclidean");
Dist= squareform(distv);

%3. Compute the WSPP;
[UOtt2,bOtt2, aOtt2, fOtt2,iterOtt2]=WSPP(Dist, 2, 50);
[pf2,Dw2,Db2] = psF(Z,UOtt2)
[UOtt3,bOtt3, aOtt3, fOtt3,iterOtt3]=WSPP(Dist, 3, 50);
[pf3,Dw3,Db3] = psF(Z,UOtt3)
sum(UOtt2)
sum(UOtt3)
[UOtt4,bOtt4, aOtt4, fOtt4,iterOtt4]=WSPP(Dist, 4, 50);
[pf4,Dw4,Db4] = psF(Z,UOtt4)
sum(UOtt4)

%4. Compute the WSP;
[UOtt2,DbOtt2, DwOtt2, fOtt2,iterOtt2]=WSP(Dist, 2, 50);
[pf2,Dw2,Db2] = psF(Z,UOtt2)
[UOtt3,DbOtt3, DwOtt3, fOtt3,iterOtt3]=WSP(Dist, 3, 50);
[pf3,Dw3,Db3] = psF(Z,UOtt3)
[UOtt4,DbOtt4, DwOtt4, fOtt4,iterOtt4]=WSP(Dist, 4, 50);
[pf4,Dw4,Db4] = psF(Z,UOtt4)

%5. Compute the PD;
[UOtt2,DbOtt2, DwOtt2, fOtt2,iterOtt2]=PD(Dist, 2, 50);
[UOtt3,DbOtt3, DwOtt3, fOtt3,iterOtt3]=PD(Dist, 3, 50);
[UOtt4,DbOtt4, DwOtt4, fOtt4,iterOtt4]=PD(Dist, 4, 10);
[pf2,Dw2,Db2] = psF(Z,UOtt2);
[pf3,Dw3,Db3] = psF(Z,UOtt3);
[pf4,Dw4,Db4] = psF(Z,UOtt4);
Q2 = UOtt2*DbOtt2*UOtt2'+UOtt2*DwOtt2*UOtt2'-diag(diag(UOtt2*DwOtt2*UOtt2'));
```

```
Q3 = UOtt3*DbOtt3*UOtt3'+UOtt3*DwOtt3*UOtt3'-diag(diag(UOtt3*DwOtt3*UOtt3'));
Q4 = UOtt4*DbOtt4*UOtt4'+UOtt4*DwOtt4*UOtt4'-diag(diag(UOtt4*DwOtt4*UOtt4'));
%verify ultramtric
verultrametrica(Q2)
verultrametrica(Q3)
verultrametrica(Q4)

Z3 = linkage(Q3);
figure();
dendrogram(Z3,0)

%2. Compute PCA to determine the number of Principal Components;
[A,L]=eigs(Sz,9);
A2=A(:,1:2);
A2r=rotatefactors(A2);

%3. Compute K-means on the number of component identified in the step 1;
Y=Z*A2r;
[vc2,Ym2]=kmeans(Y,2,'Replicates',100);
[vc3,Ym3]=kmeans(Y,3,'Replicates',100);

%4. Identify the best K with pF;
I2=eye(2)
U2=I2(vc2,:);
I3=eye(3)
U3=I3(vc3,:);
[pf, Dw, Db]=psF(Y, U2);
psF2=psF(Y,U2)
[pf, Dw, Db]=psF(Y, U3);
psF3=psF(Y,U3)

silhouette(Y,vc2);
s2=silhouette(Y,vc2)
mean(s2)
silhouette(Y,vc3)
s3=silhouette(Y,vc3)
mean(s3)

figure;
gscatter(Y(:,1), Y(:,2), vc2, 'rgb', 'o^s', 8);
hold on;
plot(Ym2(:,1), Ym2(:,2), 'kx', 'MarkerSize', 12, 'LineWidth', 2);
xlabel('PC1');
ylabel('PC2');
title('Tandem Analysis');
legend('Cluster 1', 'Cluster 2', 'Centroids');
```

```
grid on;
hold off;

%5. Compute Reduced K-mean;
[Urkm,Arkm, Yrkm,frkm,inrkm]=REDKM(Z, 2, 2, 20);
pf2=psF(Yrkm,Urkm)
[Urkm3,Arkm3, Yrkm3,frkm3,inrkm3]=REDKM(Z, 2, 3, 20);
pf3=psF(Yrkm3,Urkm3)
U2'*Urkm  %
U3'*Urkm3

figure;
gscatter(Y(:,1), Y(:,2), Urkm,'rgb', 'o^s', 8);
hold on;
title('Reduced K-Means Clustering');
legend('Cluster 1', 'Cluster 2');
grid on;
hold off;

%6. Compute Factorial K-means;
[Ufkm,Afkm, Yfkm,ffkm,infkm]=FKM(Z, 2, 2, 20);
U2'*Ufkm
[Ufkm3,Afkm3, Yfkm3,ffkm3,infkm3]=FKM(Z, 2, 3, 20);
U3'*Ufkm
pf2FK=psF(Yfkm,Ufkm)
pf3FK=psF(Yfkm3,Ufkm3)

[~, cluster_idx] = max(Ufkm, [], 2);
figure;
hold on;
colors = {'r', 'b', 'g'};
markers = {'o', 's', 'd'};
for k = 1:max(cluster_idx)
    idx = cluster_idx == k;
    scatter(Yfkm(idx, 1), Yfkm(idx, 2), 50, colors{k}, markers{k}, 'filled', 'DisplayName',
    text(Yfkm(idx, 1), Yfkm(idx, 2), num2str(k), 'HorizontalAlignment', 'center', 'Vertical
end
xlabel('First Dimension');
ylabel('Second Dimension');
title('Classification of Objects on First Two Dimensions of Factorial K-means');
legend show;
grid on;
hold off;

%7. Compute Clustering and Disjoint PCA;
[Vcdpca,Ucdpca,Acdpca, Ycdpca,fcdpca,incdpca]=CDPCA(Z, 2, 2, 20);
```

```
[Vcdpca3,Ucdpca3,Acdpca3, Ycdpca3,fcdpca3,incdpca3]=CDPCA(Z, 2, 3, 20);
pf2CDP=psF(Ycdpca,Ucdpca)
pf3CDP=psF(Ycdpca3,Ucdpa3)
[Vdpca,Adpca, Ydpca, fdpca, indpca] = DPCA(Z,2,20)

% Create a heatmap of the loadings matrix
figure;
h = heatmap(Acdpca);
h.Title = 'Loadings Matrix';
h.XLabel = 'Components';
h.YLabel = 'Variables';
h.Colormap = winter;
h.ColorLimits = [-1, 1];

%%%%%%%
explainedVarianceDim1 = 62.19;
explainedVarianceDim2 = 14.75;
Dim1 = Ycdpca(:, 1);
Dim2 = Ycdpca(:, 2);
figure;
hold on;
scatter(Dim1(group1), Dim2(group1), 50, 'b', 'o', 'filled');
scatter(Dim1(group2), Dim2(group2), 50, 'r', '^', 'filled');
ellipse(mean(Dim1(group1)), mean(Dim2(group1)), std(Dim1(group1)), std(Dim2(group1)), 'b');
ellipse(mean(Dim1(group2)), mean(Dim2(group2)), std(Dim1(group2)), std(Dim2(group2)), 'r');
xlabel(['Dim 1 (' num2str(explainedVarianceDim1) '% Variance)']);
ylabel(['Dim 2 (' num2str(explainedVarianceDim2) '% Variance)']);
title('Clustering and Disjoint PCA');
legend('Cluster 1', 'Cluster 2');
grid on;
axis equal;
hold off;

%8. Compute the Double K-means.
[Vdkm,Udkm,Ymdkm, fdkm,indkm]=DKM(Z, 2, 2, 20);
[Vdkm3,Udkm3,Ymdkm3, fdkm3,indkm3]=DKM(Z, 2, 3, 20);

%9. Compare the results by using the confusion matrix (contingency table);
%confusion matrix between 4 and 5
U2'*Urkm
%confusion matrix between 4 and 6
U2'*Ufkm
%confusion matrix between 4 and 7
U2'*Ucdpca
```